USN. | | | | | | | | | | | [24MTAPTE111]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY- 2025

GAS DYNAMICS

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

		Questions	CO's	Blooms	Marks
1.	a.	Derive all the isentropic relations for compressible gas flow.	CO1	L3	10
	b.	Determine the divergence and curl of the following vectors: i) $F= x^2y\,\mathbf{i} +(3x-z^3)\,\mathbf{j}+4y^2\,\mathbf{k}$ ii) $F= (3x+2z^2)\,\mathbf{i} + (x^3y^2/z)\,\mathbf{j} - (z-7x)\,\mathbf{k}$	CO1	L3	06
	c.	Differentiate between calorically perfect gas and thermally perfect gas.	CO1	L2	04
2.	a.	Derive the area velocity relation $$\frac{dA}{A} = (M^2 - 1)\frac{du}{u}$$ Explain the inferences from this relation.	CO2	L3	10
	b.	Derive an expression for downstream Mach number M_2 in terms of upstream Mach number M_1 for the flow of a calorically perfect gas through a normal shock.	CO2	L3	10
3.	a.	A supersonic flow with $M_1 = 2$, $P_1 = 1$ atm and $T_1 = 288$ K is compressed around a sharp corner through a deflection angle of 20 degrees. Calculate M_2, P_2, T_2, P_{02}, and T_{02} behind the shock wave.	CO3	L3	10
	b.	Air flowing at 32 kPa, 240 K, and $Ma_1 = 3.6$ is forced to undergo an expansion turn of 15°. Determine the Mach number, pressure, and temperature of air after the expansion.	CO4	L3	10
		OR			
4.	a.	Air having an initial Mach number $M_1 = 3$, a free stream static pressure $p = 1.0133 \times 10^5\ N/m^2$ and an initial static temperature 300 K is deflected through 15° angle by a frictionless surface. Assuming that a weak oblique shock wave occurs, calculate (a) the static pressure (b) the density (c) the static temperature (d) the change in stagnation pressure (e) the velocity and (f) the Mach number behind the shock.	CO3	L3	10

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	b.	A uniform supersonic flow at M_1=2.0, P_1=0.8 atm and T_1=270 K expands through two convex corners of 10 degrees each arranged one by one in a plane. Determine the downstream properties M_3, P_3, T_3 and the angle of the second expansion fan.	**CO4**	**L3**	**10**
5.	**a.**	What are the steps involved in the determination of steady two-dimensional steady supersonic flow field by the method of characteristics. Write one or two sentences of each.	**CO5**	**L3**	**10**
	b.	In the method of characteristics, explain unit processes for inside of flowfield and wall points with respect to a supersonic, two-dimensional, inviscid and irrotational flowfield.	**CO5**	**L3**	**10**
		OR			
6.	**a.**	Consider the method of characteristics applied to steady inviscid irrotational 2-D supersonic flowfield. What does the equations for the characteristic curves signify? Illustrate with a sketch (Derivation not required).	**CO5**	**L3**	**10**
	b.	Briefly explain compatibility equations valid on the respective characteristic lines, referred in 6a.	**CO5**	**L3**	**10**
7.	**a.**	It is required to measure Mach number in test section of a supersonic wind tunnel. Sketch the probe used and draw the streamlines to illustrate the flow field upstream of the probe and surrounding the probe. What additional information is required other than probe reading to calculate the Mach number. Write the equation used to calculate the Mach number from the probe reading.	**CO6**	**L3**	**10**
	b.	A K-type thermocouple is used to measure the temperature of air flowing in duct. Reference junction temperature is 25^0 C. If the voltmeter reads 2.474 mV, what is the temperature of the air flowing in the duct. Use the thermocouple data given the Table 7b. **Table 7b: Voltage E in Millivolts versus Temperature T_m for K-Type Thermocouple referenced to zero degree centigrade**	**CO6**	**L3**	**10**

			Type K						
		°C	0	5	10	15	20		
		−200	−5.891	−5.813	−5.730	−5.642	−5.550		
		−175	−5.454	−5.354	−5.250	−5.141	−5.029		
		−150	−4.913	−4.793	−4.669	−4.542	−4.411		
		−125	−4.276	−4.138	−3.997	−3.852	−3.705		
		−100	−3.554	−3.400	−3.243	−3.083	−2.920		
		−75	−2.755	−2.587	−2.416	−2.243	−2.067		
		−50	−1.889	−1.709	−1.527	−1.343	−1.156		
		−25	−0.968	−0.778	−0.586	−0.392	−0.197		
		0	0.000	0.198	0.397	0.597	0.798		
		25	1.000	1.203	1.407	1.612	1.817		
		50	2.023	2.230	2.437	2.644	2.851		
		75	3.059	3.267	3.474	3.682	3.889		
					OR				
8.	**a.**	Draw a neat sketch to illustrate the construction of a typical highspeed stagnation temperature probe in compressible flow and label the parts. Explain two design considerations and how they are addressed in the design.						**CO6**	**L3**	**10**
	b.	It is required to measure dynamic pressure of air flow at standard atmospheric condition, in a circular pipe (100 mm diameter) at a given location. Based on continuity equation it is found that average velocity of air in the pipe is 80 m/s. Illustrate with a sketch an arrangement to measure the maximum dynamic pressure. Write the governing equation used to calculate the dynamic pressure. What is the assumption made?						**CO6**	**L3**	**10**